SUPPLY CHAIN ANALYTICS PROJECT

Delivery Operations Analysis, Bottleneck Detection & Predictive Modeling

Dataset

DataCo Supply Chain Dataset

Total Records Analyzed

180,519 Orders across Multiple Regions & Departments

Tools & Technologies

Python | Pandas | Seaborn | Matplotlib | Scikit-learn | SMOTE

Prepared for Interview Presentation

1. Executive Summary

This project tackles a real-world business problem faced by a global e-commerce company: too many orders are being delivered late, and that is costing both customers and profits. The company needed a data-driven way to understand why delays happen, where they happen most, and how to predict them before they occur.

Think of it like this — imagine you run an online store and more than half of your deliveries are running late. Customers are unhappy, you are losing money, and you don't even know where to start fixing it. This project provides that starting point.

What Was Done

- Analyzed 172,765 active orders (after removing cancelled ones) across multiple regions, departments, and shipping modes
- Calculated exactly how many days each order was delayed versus its scheduled delivery date
- Identified which regions, shipping modes, customer segments, and departments had the highest delay rates
- Performed root cause analysis to find the key drivers of late deliveries in the worst-performing regions
- Built a Machine Learning model (Random Forest) to predict whether an order will be late — before it happens

Key Findings at a Glance

172,765 Total Orders Analyzed	**94,523** Late Deliveries
54.7% Late Delivery Rate	**45.3%** On-Time Delivery Rate
$7.5M Total Profit (Profitable Orders)	**$2.05M** Profit Tied Up in Delayed Orders

The headline number: over half of all orders are delivered late. This is not a small inefficiency — it is a systemic problem that the analysis has diagnosed in detail.

2. Problem Statement & Dataset Overview

The Business Problem

A global e-commerce company manages end-to-end order fulfillment across multiple regions. Products span categories like Sporting Goods, Health & Beauty, Technology, and more. The company was experiencing a critical issue: actual delivery times were frequently missing their scheduled targets, leading to:

- Customer dissatisfaction from late or unpredictable deliveries
- Revenue leakage as delayed orders are correlated with lower profitability
- No early warning system to flag high-risk orders before they go out

The Dataset

The DataCo Supply Chain Dataset is a comprehensive, real-world supply chain dataset containing order-level details. Here is what it covers:

Attribute	Details
Total Raw Records	180,519 orders
Orders Used for Analysis	172,765 (after removing cancelled orders)
Original Features	53 columns
Features After Cleaning	~20 relevant features
Time Period	Multi-year order history
Geographic Coverage	Multiple global regions including Africa, Americas, Asia, Europe
Customer Segments	Consumer, Corporate, Home Office
Shipping Modes	Standard Class, Second Class, First Class, Same Day

Data Cleaning Steps

Before any analysis, the data was cleaned systematically:

- Dropped 33 irrelevant columns (customer PII, product images, order IDs, etc.) that had no impact on delivery analysis
- Removed 7,754 cancelled orders — these don't represent real delivery performance
- Fixed date format inconsistencies and converted date strings to proper datetime objects
- Verified zero duplicate records and zero missing values in key columns

New calculated columns created:

- Order Processing Time: Actual days from order placement to shipment
- Delay: Actual processing time minus the scheduled shipment days (positive = late)
- Is_Delayed: A simple True/False flag for whether an order was late
- Order_Month, Order_Day, Order_Hour: Time-based features for pattern analysis
- Profitability Flag: Whether each order resulted in Profit, Loss, or Break-even

3. Delivery Operations Analysis

Understanding Delays

The first step was to calculate how delayed each order actually was. Delay was defined as:

$$\textit{Delay} = \textit{Actual Processing Time} - \textit{Scheduled Shipment Days}$$

A positive Delay value means the order shipped later than promised. A negative value means it shipped early (an advance shipment).

Overall Delay Distribution

The distribution of delays across all 172,765 orders reveals a clear pattern: most orders are either on time (0 delay) or delayed by 1 to 4 days. Very few orders arrive extremely early or extremely late — making the 1-4 day delay window the primary target for optimization.

Key observation: The 90th percentile delay is approximately 4 days, meaning 90% of late orders are delayed by 4 days or fewer. This is an actionable window — a relatively small improvement in scheduling accuracy could recover a large portion of on-time deliveries.

Profitability Distribution

Not all orders are equally profitable, and delay has a direct financial impact:

Profitability Category	Approximate Share	Financial Implication
Profitable Orders	~60%	Generate $7.5M in total profit
Loss-Making Orders	~35%	Erode margins; overlap heavily with delayed shipments
Break-even Orders	~5%	Neutral impact; volume-driven

Profitability vs. Delivery Time

A critical insight emerged when crossing delay data with profit data: orders that arrive on time (0 delay) tend to have higher average profit per order. As delay increases, mean profit drops. This is not necessarily because delays cause lower profit directly — rather, delayed orders are often associated with less desirable shipping modes, regions, or order types that have structural profitability challenges.

The $2.05 million figure tied up in delayed orders represents the total profit across all orders that were delayed. This does not mean all that profit was lost — but it shows the financial exposure created when the system fails to deliver on time.

4. Bottleneck Detection

Bottleneck detection answers: 'Where exactly is the system breaking down?' Rather than looking at the overall 54.7% delay rate, the analysis breaks it down by every meaningful category to pinpoint the worst offenders.

Delay Rate by Region

Certain geographic regions consistently perform worse than others. The top 5 regions by delay percentage are:

Region	Delay %	Risk Level
Central Africa	58.7%	Critical
Southeast Asia	56.0%	High
Eastern Europe	55.9%	High
South of USA	55.9%	High
East Africa	55.9%	High

Delay Rate by Shipping Mode

This is one of the most actionable findings. The shipping mode selected for an order has a massive impact on whether it will be late:

- Standard Class has the highest delay rate — this is the cheapest and slowest option, yet it is used for 62% of all orders
- Same Day delivery, despite being the fastest, still shows significant delays — suggesting fulfillment capacity issues on urgent orders
- First Class and Second Class fall in between, but all modes show delay rates above 40%, indicating the problem is systemic, not isolated to one tier

What this means: Simply upgrading customers to a faster shipping tier is not sufficient. The root cause lies in order processing and fulfillment, not just the shipping carrier.

Delay Rate by Customer Segment

Customer Segment	Delay %	Business Impact
Consumer	~55%	Largest segment — affects most customers

Customer Segment	Delay %	Business Impact
Corporate	~54%	B2B clients — reputation risk if SLAs missed
Home Office	~54%	Smallest segment but similar delay exposure

Delay Rate by Department

Some departments have structurally higher delay rates, which likely reflects the weight, sourcing complexity, or popularity of products in those categories:

Department	Delay %	Rank
Health and Beauty	56.9%	#1 Worst
Pet Shop	56.6%	#2
Technology	55.7%	#3
Outdoors	55.6%	#4
Fitness	55.4%	#5

5. Root Cause Analysis

Bottleneck detection tells us where delays are highest. Root cause analysis goes one layer deeper and asks: within the worst-performing regions, what specific factors are driving the delays?

A custom function was built to analyze the top delay drivers for any given region across five dimensions: Shipping Mode, Customer Segment, Department Name, Payment Type (Type), and Order Status. Let's walk through the two most problematic regions.

Central Africa — Top Delay Drivers

Central Africa has the highest delay rate at 58.7%. The root cause breakdown reveals:

- Standard Class shipments to Central Africa have the highest delay rate of any mode-region combination — the slow shipping tier combined with long-distance logistics creates a compounding effect
- Orders from the Consumer segment dominate volume here, amplifying the impact of any inefficiency
- Technology department orders show particularly high delays — likely due to customs checks, handling requirements, or sourcing lead times for electronics
- Debit payment type orders correlate with higher delays (potentially due to payment verification processing time)
- Orders already in 'Processing' or 'Pending' status show predictably high delay rates — these are the orders where early intervention would have the highest impact

East Africa — Top Delay Drivers

East Africa shows a very similar pattern to Central Africa, with 55.9% delay rate:

- Standard Class is again the primary culprit — same structural problem, different geography
- Health and Beauty department orders are particularly problematic here, potentially due to import regulations or cold-chain requirements
- Corporate segment orders in East Africa face significant delays, which is a serious concern for B2B relationships and contract SLAs

Cross-Regional Pattern

The pattern is consistent across regions: Standard Class shipping + specific high-risk departments + long-distance geographies = near-guaranteed delay. This is a strong signal that the fix is not region-specific — it requires a systemic improvement in how shipping mode is matched to geography and product type.

6. Time-Based Delay Patterns

Delays don't happen randomly across time. By analyzing when orders are placed, we can identify predictable peak-risk windows — times when the fulfillment system is most likely to struggle.

Monthly Patterns

Delay rates fluctuate across months of the year. The analysis identified the top 3 highest-delay months, which tend to coincide with:

- Peak shopping seasons (e.g., holiday months) where order volumes surge and fulfillment capacity gets stretched
- Months with known supply chain disruptions or seasonal product demand spikes
- End-of-quarter periods where bulk corporate orders may flood the system

Practical implication: The company should increase fulfillment staffing, pre-position inventory, and consider restricting certain slow shipping options during peak delay months.

Day of Week Patterns

Delay rates vary meaningfully by the day an order is placed:

- Weekend orders (Saturday and Sunday) tend to have higher delay rates — this makes sense because orders placed on weekends often cannot be processed until Monday, adding 1-2 days to effective processing time
- Monday and Friday also show elevated delay rates — Monday due to the backlog of weekend orders, and Friday because orders placed at end of business often miss the daily cutoff
- Mid-week orders (Tuesday through Thursday) tend to have the best on-time performance

Practical implication: The company could incentivize mid-week ordering with small discounts, or set more accurate expected delivery dates for weekend orders to manage customer expectations.

Hour of Day Patterns

Even the hour of day when an order is placed affects delay probability:

- Orders placed after business hours (evenings and late nights) show higher delay rates — these orders sit in queue overnight

- Early morning orders (9am to 11am) tend to have the lowest delay rates — they enter the fulfillment pipeline at the start of the workday
- The difference between best and worst hours can be 5-8 percentage points in delay rate

Practical implication: Setting realistic cutoff times for same-day processing and communicating these clearly to customers could reduce both delays and customer complaints.

7. Machine Learning Model — Predicting Late Deliveries

This is where the analysis moves from 'understanding the past' to 'predicting the future.' The goal: build a model that can look at an order when it is placed and flag it as high-risk for late delivery — before it actually becomes late.

Why Prediction Matters

Knowing an order will likely be late lets the company:

- Alert the customer proactively and set accurate expectations
- Route the order to a faster shipping mode before it ships
- Prioritize the order in the fulfillment queue
- Trigger a manual review for high-value orders at risk

Target Variable

Target	Value	Count
Late_delivery_risk	1 (Will be Late)	98,977 orders
Late_delivery_risk	0 (Will be On Time)	81,542 orders

Features Used for Prediction

The model was trained on 9 features that are available at the time an order is placed (no information leakage):

Feature	Why It Matters
Type (Payment Method)	Debit, Transfer, Cash — different processing times
Days for Shipment (Scheduled)	Promised delivery window — shorter windows = higher risk
Category Name	Product type affects handling and sourcing time
Customer Segment	Consumer vs Corporate vs Home Office priority handling
Department Name	Some departments have structural delay issues
Order Region	Geography is one of the strongest delay predictors
Shipping Mode	The single most impactful shipping decision
Order Month	Captures seasonal demand peaks
Order Hour	Time of day affects same-day processing cutoff

Feature Engineering — Frequency Encoding

Categorical features like Region and Department were converted to numbers using Frequency Encoding. This means each category is replaced by how often it appears in the dataset (as a proportion). For example, if Standard Class appears in 62% of orders, it gets encoded as 0.62.

Why this approach? It is lightweight, handles high-cardinality categories well, and preserves relative importance information without creating hundreds of dummy columns (which would slow the model and create noise).

Handling Class Imbalance — SMOTE

The dataset had more late deliveries (98,977) than on-time deliveries (81,542). While not severely imbalanced, applying SMOTE (Synthetic Minority Over-sampling Technique) on the training data helped the model learn patterns for both classes equally.

What is SMOTE? It creates synthetic examples of the minority class by interpolating between existing examples — essentially teaching the model with more balanced experience. After SMOTE, the training set had equal counts for both classes.

Model — Random Forest Classifier

A Random Forest was chosen for this task. Think of it as building hundreds of different decision trees (each trained on a slightly different subset of the data), and then taking a majority vote across all trees for the final prediction.

Why Random Forest?

- Handles mixed feature types (numbers and categories) naturally
- Robust to outliers and noisy data
- Built-in feature importance ranking
- No need for feature scaling
- Industry-standard baseline for tabular classification problems

Model Performance

Metric	Score	What It Means
Accuracy	High	Overall % of orders correctly classified
Precision	High	Of predicted 'Late' orders, how many actually were late
Recall	High	Of all actual late orders, how many did the model catch

The model achieved strong performance on all three metrics. In a supply chain context, Recall is especially important: missing a late order (false negative) means a customer gets a surprise delay. High recall means the model catches most at-risk orders.

8. Business Recommendations

Based on all the analysis, here are concrete, actionable recommendations the company should implement:

1. Deploy the Predictive Model in Production

Integrate the trained model into the order management system. Every time a new order is placed, run it through the model. For orders flagged as high-risk:

- Send an automated alert to the fulfillment team to expedite processing
- Offer the customer a proactive notification with a revised estimated delivery date
- Auto-suggest a faster shipping upgrade if the margin on the order justifies the cost

2. Fix the Standard Class Problem

Standard Class is used for 62% of all orders and consistently shows the highest delay rate. The company should:

- Audit Standard Class carrier performance by region and renegotiate SLAs with underperforming carriers
- Identify which routes are worst and pre-route those orders through a more reliable carrier
- Consider geo-fencing certain high-risk region + Standard Class combinations — automatically upgrading to Second Class

3. Focus Operations on the Top 5 Delay Regions

Central Africa, Southeast Asia, Eastern Europe, South USA, and East Africa account for a disproportionate share of late deliveries. For these regions:

- Establish regional fulfillment hubs or partnerships to reduce last-mile distance
- Create region-specific shipping playbooks defining which products should use which shipping modes
- Set more conservative promised delivery dates to build in buffer time

4. Address Seasonal and Temporal Peaks

Since delay rates spike in certain months, days, and hours:

- Increase fulfillment staffing during peak delay months (identified from the monthly trend analysis)
- Implement order cutoff times visible to customers: 'Order by 11am for same-day processing'
- Consider a 'Weekend Delivery Guarantee' program to address the known weekend delay spike

5. Monitor High-Risk Departments

Health & Beauty, Pet Shop, and Technology departments have the highest delay rates. These departments likely face supply chain complexity that the general fulfillment pipeline wasn't designed for:

- Create dedicated fulfillment lanes for these departments
- Track on-time performance by department in a real-time dashboard
- Work with procurement to ensure adequate safety stock for high-velocity, delay-prone SKUs

9. Project Architecture & Summary

End-to-End Workflow

Step	What Was Done
1	Data Loading & Exploration — Loaded 180,519 records, examined all 53 columns, understood data types and distributions
2	Data Cleaning — Dropped 33 irrelevant columns, removed 7,754 cancelled orders, fixed date formats
3	Feature Engineering — Created Delay, Is_Delayed, Order Processing Time, time features, and Profitability Flag
4	KPI Dashboard — Computed business metrics: 54.7% late rate, $7.5M profit, $2.05M at-risk from delays
5	Profitability Analysis — Crossed delay data with profit to understand financial impact
6	Bottleneck Detection — Computed delay % across 6 categories to find worst-performing segments
7	Root Cause Analysis — Built a reusable function to drill into top delay drivers per region
8	Time-Based Analysis — Analyzed delay patterns by month, day of week, and hour of day
9	ML Modeling — Frequency encoded features, applied SMOTE, trained Random Forest, evaluated on held-out test set

Technologies & Libraries

Library	Purpose
Pandas	Data loading, cleaning, transformation, and aggregation
NumPy	Numerical operations and vectorized calculations
Matplotlib	Custom multi-panel visualizations and chart formatting
Seaborn	Statistical bar plots, pair plots, and palettes
Scikit-learn	Train/test split, Random Forest model, accuracy/precision/recall metrics
Imbalanced-learn (SMOTE)	Synthetic oversampling to balance training data

Key Takeaways

- This is not just an EDA project — it delivers a working predictive system that has real operational value
- Every step has a business justification: cleaning decisions, feature choices, and model selection are all tied to the problem statement
- The analysis is structured from 'What is happening?' → 'Where is it happening?' → 'Why is it happening?' → 'Can we predict it?'
- The insights are specific and actionable — not generic observations but region-level and department-level findings the operations team can act on
- SMOTE and frequency encoding demonstrate awareness of real ML engineering challenges (imbalanced data, categorical features)

End of Report